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Deep Regression hybridized Neural Network in human stress detection

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***Abstract*—In today's environment, psychological problems such as stress, sadness, and anxiety become significant. Millions of individuals suffer from depression every year, but only a least percentage of them undergo timely treatment. As a result, it becomes critical to identify individual tension and means to relax instantly. It is necessary to recognize and handle stress before it becomes a severe problem. The primary goal is to use Machine Learning techniques to detect anxiety in people to improve their quality of life. The vast majority of stress datasets have been collected while consumers were subjected to high-stress situations, such as music, video files, significant equipment failure, visual information, and gameplay situations. However, whether certain databases are used to construct models that can accurately identify anxiety in various situations is still an open question. This paper offers a hybrid strategy DR-CNN for multimodal stress detection that uses a convolutional neural network (CNN) and a regression classifier. Our findings show that this technique has promise, but they also highlight the need for more research to understand better the relevance of deep-learning approaches in designing generally applicable designs for multidimensional emotion detection.**

Keywords—mental states, convolutional neural network (CNN); machine learning; deep-learning

I. Introduction

Stress has been ingrained in our daily lives and is a well-known global health concept. In today's competitive market, anxiety has only lately become an accepted component of professional life. An employee must constantly deal with various conditions at work, job instability, including work overload and a lack of employment contentment as shown in Fig.1. Stress can cause a variety of severe health impacts, including hypertension, lack of sleep, illness susceptibility, and coronary heart disease.

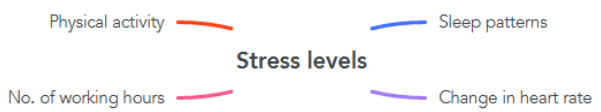

Fig. 1. Conceptual Model of Stress

These circumstances lead to mental stress, which is now the primary cause of various disorders. Such negative consequences affect not just individuals' health and quite well but also professional profit and productivity. Having the ability to recognize stress can assist individuals in identifying a variety of stress-relieving options before unfavorable outcomes occur as shown in Fig.2. Psychology surveys [1] or consultation [2] are used in traditional stress detection procedures. The stress measure is highly dependent on the individual's impression because the survey is heavily reliant on people's opinions. The resulting score would indeed be one-sided when people chose to disclose their states of mind with reservations.

Fig. 2. Types of Stress

Psychological stress, episodic acute stress, and acute stress are the three forms of stress based on the period of stress. Acute stress occurs for a brief period, but chronic stress occurs over longer period [3]. To be healthy and maintain the performance of its organs, the human body requires a certain degree of stress. On the other hand, unmanaged stress can be harmful to one's health and increase the chance of significant long-term difficulties. Obesity, heart attacks, prediabetes, stomach aches, body aches, prostatic illness in men, and interruption of a woman's menstrual period are all symptoms of chronic stress. As a result, maintaining a healthy, tense existence takes a lot of self and appropriate coping strategies to deal with stressors. When the human body is stressed, it triggers a "fight or flight" reaction to protect itself from numerous threats. Adrenaline and cortisol chemicals are released into the bloodstream in this response to asking the body to respond rapidly.

There are many publicly available hormonal datasets for psychological science [4-8] that were analyzed emotionally in a frame of reference. Clients have to expose to high-stress events that happen in everyday life. Even though such advances can produce a predictive stress model with high classification results, it's unclear if they'll help detect subtle stress, which is common in situations like UX evaluation studies.

Artificial intelligence has recently been acknowledged as having a function in recognizing depression and work-related stress [9-12]. Prosodic features, formants, spectral features, and changes of features of the voice are the most commonly employed features for stress detection. However, the difficulties of recognizing effort anxiety and depression symptoms in malware environments have not been investigated. The emotion recognition period and the

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emotion classification phase were separated in our model. After completing the emotion detection phase, we administered stress and emotion-related surveys. Finally, regression analysis is used to examine the association between emotions and visually apparent psychological stress as stated by psychologists in the survey replies, based on these surveys and the emotion detection results.

There are numerous psychology and medical technology methods, such as consulting or medicine, for dealing with stress, particularly stress that arises at work. However, all of these strategies will be useful only if it is possible to estimate a person's stress level without the subject being aware of it [13]. As a result, a stress monitoring that indicates the level of stress without asking the issue replies to questions, biological samples, or sport electrodes should be developed, as this will assist avoid interfering with a person's work schedule. This type of technology isn't meant to take the place of scientific medical procedures for determining stress levels entirely and adequately. Still, it could help determine when such operations are necessary [14-16]. This is particularly desirable because biomedical or psychiatric operations cannot be performed indefinitely, necessitating the use of indicators to aid in the assessment and treatment of patients. This article aims to propose and implement a more accurate method for identifying emotions from facial muscle movement using Convolutional Neural Networks [17] and then use these recognized emotional responses to assess facially observable anxiety levels by selecting the optimal expression stress relational database model from stress survey data. In section 2, the related work is presented. Section 3 contains information on how to execute the plan into action. Section 4 details the experiment. Finally, section 5 concludes with a conclusion and recommendations for future work.

II. Literature Survey

As a consequence of their regular work obligations, people incur psychological and physical strain. Every individual faces stress over time, and how they deal with it is likely to be influenced by the real and perceived repercussions of their actions. Coping with stress demands applying a variety of personal and situational resources [1]. Numerous studies have been conducted in this area.

Ravinder Ahuja et al. [1] focused on calculating students' mental stress one week before the exam and while using the internet. The purpose is to look at the tension in college students from different periods of their lives. Exam stress or recruiter stress have an unseen influence on students. They look at how these elements affect a student's cognition and see if there's a link between stress and time spent online.

Padmaja et al. [2] identify stress using sensor technology and a machine learning approach. Each stressor's influence was assessed separately utilizing regression analysis, and a combined model was proposed and analyzed using a continuous logit model such as logistic regression, probit, and supplemental data. In a real-time scenario, data from personnel employed inside the IT and other industries across India was used to test and evaluate this system. The work is unique because it focuses on building an anxiety detection approach for the user as non-invasive as possible.

To detect and study the impacts of stress, P. Yuen et al.[3] suggested a hybrid deep learning technique. The Vgg-16 networks take feature maps as input, which makes integrating such dual deep learning models, Resnet and Vgg-16, particularly advantageous. The raw picture is the first network's input (DNN-1), and the Fundamental features obtained by the wavelet transform are the subsequent network's input (DNN-2). The frequency characteristics aid in the separation of resonance data from the images. Finally, the findings of these two systems are integrated to improve classification accuracy. Both networks are capable of detecting stress is difficult, but the combined network outperforms, including both terms of higher productivity. Since the characteristics from both networks have been retrieved and concatenated.

Cristian Paul et al. [4] describe their first steps to build a deep-learning-based strategy to aid in diagnosing multimodal stress. The findings show that this direction has promise. Still, they also highlight the need for more research to understand better the relevance of deep-learning methodologies in designing generally applicable architectures for heterogeneous affective computing. We evaluate our approaches on eight different info signals acquired from 28 individuals using the MuSE dataset, a rich resource to study the relationships between stress and emotion.

The effectiveness of the currently accessible physiological dataset of customer experience evaluation is investigated by Liapis et al. [5]. EDA and ST signals were used to train three classic ML classifiers based on convolutional neural networks incorporating continuous variables and entity embeddings. High accuracy was attained for both instructional methods for the binary classification issue (stress vs. no stress)—the results of the generated models' stress detection effectiveness in various contexts.

I. Madhavi et al. [6] propose a unique deep learning strategy based on the Growing Self Organizing Map and CNN algorithms to overcome these issues. For noise reduction, speaker diarization and sampling strategy techniques are used in addition to this innovative Artificial Intelligence approach. Pitching and speech enrichment approaches to address the data problem of information asymmetry that most real-world datasets suffer from. A benchmark dataset is used to illustrate the effectiveness and accuracy of the suggested method (DAIC-WOZ).

Reshma et al. [7] offer a method for detecting relaxation and stress expressions on a tweeter dataset, i.e., to find emotions or feelings regarding daily life conduct sentiment analysis. An analysis is a way of extracting relevant sentiment-related data from text. We used the TensiStrength technology for emotion strength identification on social networking websites to derive sensitivity from the simple English text. TensiStrength is a method that detects the intensity of relaxation and stress indicated in text messages on media. TensiStrength sees active and passive signs of tension or calm using a linguistic technique and a set of criteria. This system categorizes both excellent and unpleasant thoughts using a strength scale of -5 to +5 to indicate sentiments. The interview's stressed phrases are analyzed and divided into stress and relaxation. TensiStrength is a versatile social web tool that may be used in a variety of situations.

The goal of Prerna et al. [8] uses Machine Learning techniques to detect stress in people, with the ultimate goal

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of enhancing their quality of life. To use a public information multidimensional dataset called WESAD, they propose various ML models for detecting stress in individuals. For three states - amusement, stress, and neutral - sensor data such as ECG, body temperature, respiration, electromyography, and electromyographic activity (EDA) is collected. Deep learning techniques such as k-NN, Randomized Forest, AdaBoost, and Svms were used. The Random Forest model beat other methods in binaries and regimen, with Exhibited strong of 83.34 and 65.73, respectively.

III. System Model

Facial expression identification and stress detection from interpreted emotions are the two primary processes in our model. For all emotions, we used a Convolution Neural Network (CNN) to calculate the odds that a facial expression represents that emotion as shown in Fig.3.

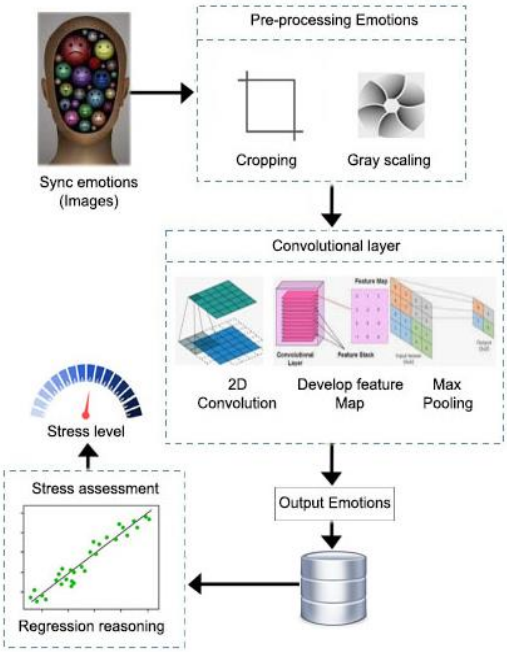

Fig. 3. Design Model

The acquired image is preprocessed in the input layer for emotion recognition. Cropping and conversion to greyscale of the pictures captured were part of the preprocessing procedure, which resulted in images with a normalized size or intensity.

After that, the picture was normalized and loaded into a NumPy array. The NumPy array was sent to the convolutional layer to build the feature maps, which consisted of three convolution layers and a set of filters. In place of completely connected layers, a combination of depth-wise separable convolutions and residual modules are used. The ADAM optimizer was used to train the model. To remove any fully connected layers, Global Average Pooling is also employed. The implementation was carried out by utilizing a softmax function for each lower-dimensional mapping in the penultimate convolution layers with the same number of layers as the number of courses. Finally, the softmax function displayed the outcome as a chance for every emotional class in the output layer as shown in Fig.4.

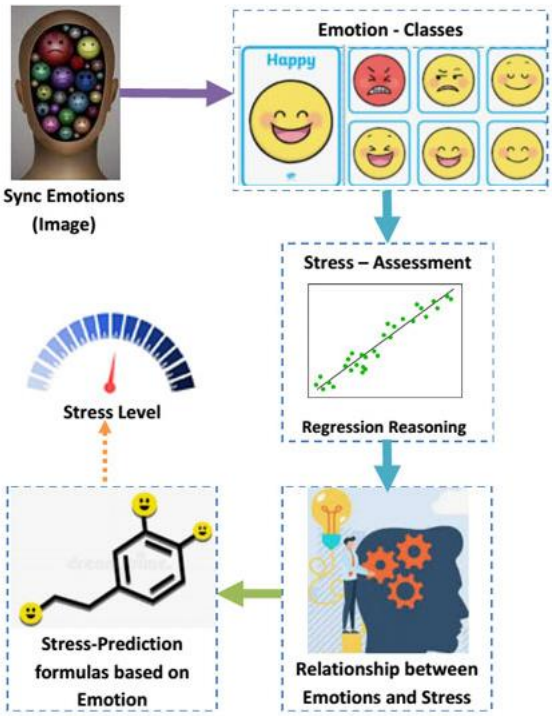

Figure 4. Model for Stress Assessment

The stress levels were analyzed after deciphering emotion information from facial expressions, and to that end, a poll was conducted among people that ties emotional levels to stress levels. The survey was carried out without the use of any images. Respondents were given percentages of emotion intensity ranging from 0 to 100 in steps of 20, and they have been asked to correlate each level of the fundamental emotion with a stress level ranging from 0 to 9. A total of 150 replies were received.

The survey findings and the decrypted emotion information data were used to forecast a regression design that accurately describes the stress and emotion data after the survey was completed. Five separate linear and non-linear multiple regressions patterns were used for predicting stress levels from emotion degrees or percentages [18]. The best model was selected based on characteristics like the goodness of fit and root mean square errors. Exponential fit was found to be the suitable model out of all of them, and it also followed the Weber-Fechner law. The logarithmic model equations were then provided by applying regression analysis to determine the coefficients of unknown factors. Finally, the approach mentioned above can be used to assess visually perceptible anxiety levels (in terms of probability) as visible on a model's face in terms of seven fundamental expressions, with the emotional states interpreted using facial muscle movement data.

CNN for Deep Regression has a benefit over the linear model. However, regression analysis can establish a direct proportionality among attributes and outputs and hence cannot learn sophisticated non-linear correlations. As a result, additional solutions must comprehend the complex quasi-relationship between the features and the goal. A few of these techniques are Deep Convolution Networks.

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DNN can understand the complicated relationship between characteristics and outcomes since each level has a kernel function.

Pseudocode: Deep Regression using CNN

```
Load: recordset
Outcome: stress[]
# find the recordset
def find_recordset():
A, b = build_regression(m_populations=1500,
m_features=20, m_informative=10, m_tarfinds=5,
random_state=4)
return A, b
 # find the pattern
def find_pattern(m_loads, m_outcomes):pattern =
Sequential()
pattern.sum(Dense(30, load_dim=m_loads,
core_initializer='he_uniform', activation='relu'))
pattern.sum(Dense(m_outcomes))
pattern.compile(loss=KIE, optimizer='ADAM')
return pattern
# Calculate a pattern using repeated k-fold cross-
validation
def Calculate_pattern(A, b):
outcomes = list()
m_loads, m_outcomes = A.shape[1], b.shape[1]
 # Define evaluation procedure
 cvr = RepeatedKFold(m_splits=10, m_repeats=3,
random_state=1)
 # Enumerate folds
for training_iA, testing_iA in cvr.split(A):
# Prepare record
A_training, A_testing = A[training_iA], A[testing_iA]
b_training, b_testing = b[training_iA], b[testing_iA]
# Define pattern
pattern = find_pattern(m_loads, m_outcomes)
# Fit pattern
pattern.fit(A_training, b_training, verbose=0, epochs=100)
# Calculate pattern on testing set
kie = pattern.Calculate(A_testing, b_testing, verbose=0)
 # Store Outcome
print('>%.3f' % KIE)
outcomes.append(KIE)
return outcomes
# Load recordset
A, b = find_recordset()
# Calculate pattern
outcomes = Calculate_pattern(A, b)
# Review performance
print('KIE: %.3f (%.3f)' % (mean(outcomes),
std(outcomes)))
# Create a prediction for new record
row = [-.99,2.19,-.42,-.21,-1.13,-.55,-.63,-.87,-.99]
newA = asarray([row])
stree = pattern.predict(newA)
print('Predicted: %s' % stress[0])
```

IV. Experimental Results

The experiment was carried out using Kaggle dataset for stess and executed in an Intel Core i5-2.30 GHz with a Windows 7 of 64 bit and Jira with a memory of 16 GM. The experiment was done in the softmax feature map. Figures 5-8 shows the results of the emotion classification test and the stress level detection task. These figures show the related image, as well as the identified emotion label, likelihood, and stress probability.

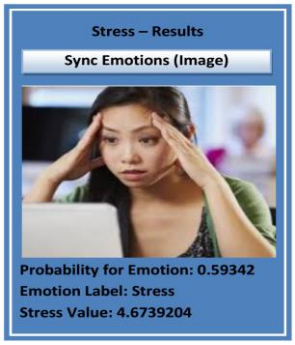

Fig. 5. Stress Output - Image of an Individual (Sample 1)

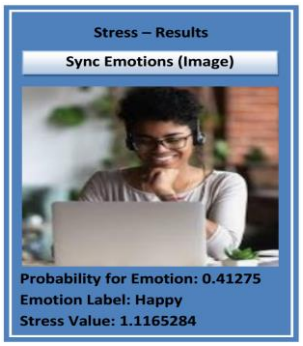

Fig. 6. Stress Output - Image of an Individual (Sample 2)

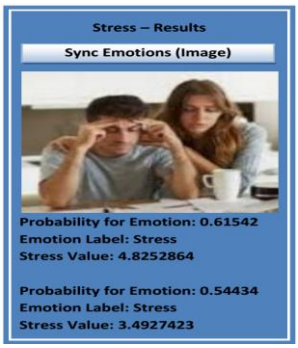

Fig. 7. Stress Output - Image of a Group (Sample 1)

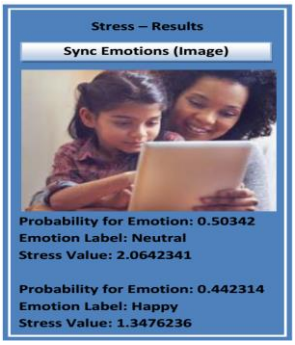

Fig. 8. Stress Output - Image of a Group (Sample 2)

The experiment tested revealed a performance of proposed DR-CNN with a F1 Score of 80.71 for a 3 class regression and 95.13 towards F1 Score in predicting a binary class problem on stress dataset.

TABLE I. Performance Evaluation

Techniques	three-class		binary	
	Fl	Accuracy	F1	Accuracy
DR	54.13	69.16	84.12	88.09
kNN	64.96	73.91	85.53	88.52
CNN	75.87	82.24	90.81	94.12
DR-CNN	80.71	85.92	95.13	96.51

DR-CNN outperforms all other classifiers in terms of total accuracy. These findings outperform earlier research, with

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accuracy rate of 80.71 percent on three statistical tests and 96.51 percent on binary class tests.

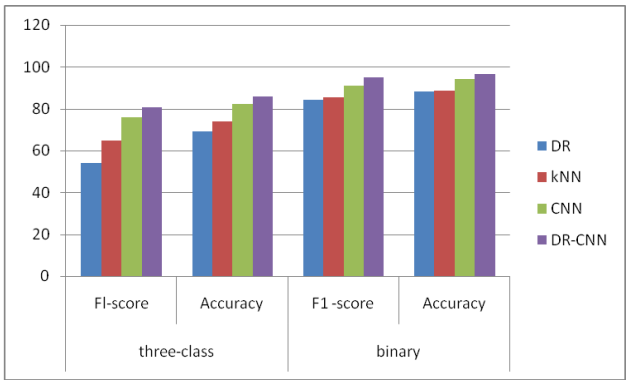

Fig. 9. Performance evaluation of stress

V. Conclusion

There are a lot of physiological datasets that were gathered in publically available stress studies. The bulk of these were taken in an environment wherein participants were subjected to high levels of stress that are common in real-life situations. Although such approaches can produce models for stress forecasting with a good accuracy rate, it's unclear if they can successfully capture subtle stress reactions common in varied situations. We provide a novel deep learning approach for detecting work-related stress in this study. We plan to verify this approach in other databases and undertake cross-database validation as part of future work. Additionally, we can improve the accuracy of our model by training it on a GPU or in the cloud. We also wish to expand our research to include real-time stress monitoring of employees in enterprises.

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